WORKED SOLUTION

STEP II 2007, Question 13

Mechanics • Birthday problem

STEP II 2007, Question 13

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/07-s2-q13

This question has a full guided treatment in *Beyond STEP*, Section 9.9, *The Birthday Problem*: https://beyondstep.keithmcnulty.org/ch09/birthday.html

The key idea

The probability that *no* two guests share a birthday is a product of factors of the form $\frac{365-r}{365}$, and products are awkward. The first part turns each factor into an exponential, and a product of exponentials is the exponential of a sum, which we can add up exactly. For the host, the guests' events are independent, so the complement is a single power, and the same approximation applies.

Solution

The approximation

For any real x,

$$e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + \cdots .$$

Put $x = r/n$. Since r is much smaller than n, x is small and the terms in x^2 and beyond are negligible compared with x. So

$$e^{-r/n} \approx 1 - \frac{r}{n} = \frac{n-r}{n},$$

as required.

At least two guests sharing a birthday

Let A be the event that at least two of the k guests share a birthday. It is easier to find $P(A')$, the probability that all k birthdays are different. Take the guests one at a time. The first can have any birthday. Given that the first r guests have r different birthdays,

the $(r+1)$th guest must avoid those r days, which happens with probability $\frac{365-r}{365}$, because the birthdays are independent and uniform over the 365 days. Multiplying,

$$\mathrm{P}(A') = \prod_{r=0}^{k-1} \frac{365-r}{365}.$$

For a party of modest size, every r here is much smaller than 365, so by the first part, with $n = 365$,

$$\begin{aligned}\mathrm{P}(A') &\approx \prod_{r=0}^{k-1} \mathrm{e}^{-r/365} = \exp\Big(-\frac{1}{365}\sum_{r=0}^{k-1} r\Big) \\ &= \exp\Big(-\frac{1}{365}\cdot\frac{k(k-1)}{2}\Big) = \mathrm{e}^{-k(k-1)/730}.\end{aligned}$$

(The $r = 0$ factor is exactly $1 = \mathrm{e}^0$, so including it costs nothing.) Hence

$$\mathrm{P}(A) = 1 - \mathrm{P}(A') \approx 1 - \mathrm{e}^{-k(k-1)/730},$$

which is the required result.

Ask yourself Isn't it worrying that we approximate many factors at once? The errors might pile up.

Each factor's error is of order $(r/365)^2$, while the exponent itself is of order $k^2/730$. For the values of k we need (around 20), the largest $r/365$ is about 0.06, and the total relative error in the exponent is only a few per cent. The approximation gives 0.500 at $k = 23$ against a true value of 0.507.

Smallest party for an even chance

We need

$$1 - \mathrm{e}^{-k(k-1)/730} \geq \tfrac{1}{2} \iff \mathrm{e}^{-k(k-1)/730} \leq \tfrac{1}{2} \iff \frac{k(k-1)}{730} \geq \ln 2,$$

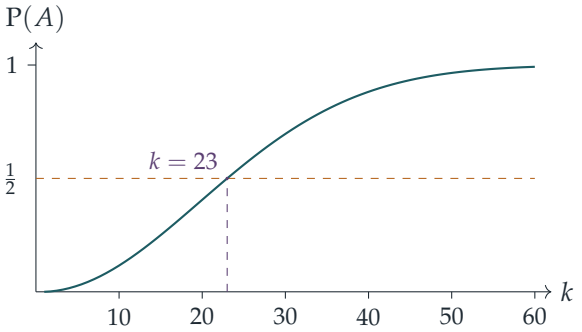
taking logarithms (the logarithm is increasing, so the inequality keeps its direction) and multiplying by -1 (which reverses it). Using $\ln 2 \approx \frac{253}{365}$, the condition becomes

$$k(k-1) \geq 730 \times \frac{253}{365} = 2 \times 253 = 506.$$

Since $k(k-1)$ increases with k for $k \geq 1$, and

$$22 \times 21 = 462 < 506, \qquad 23 \times 22 = 506,$$

the smallest value is $k = 23$.



Sharing the host's birthday

Now let m be the number of guests (the host is not counted among them). Let B be the event that at least one guest has the same birthday as the host. Whatever the host's birthday is, each guest independently misses it with probability $\frac{364}{365}$, so

$$\mathrm{P}(B') = \left(\frac{364}{365}\right)^m \approx \left(\mathrm{e}^{-1/365}\right)^m = \mathrm{e}^{-m/365},$$

using the first part with $n = 365$ and $r = 1$. This time $r/n = 1/365$ really is small, whatever the size of m.

We need $\mathrm{P}(B) \geq \frac{1}{2}$, that is $\mathrm{e}^{-m/365} \leq \frac{1}{2}$. Exactly as before this is

$$\frac{m}{365} \geq \ln 2 \approx \frac{253}{365} \iff m \geq 253.$$

So there must be at least 253 guests.

? **Ask yourself** Why is the answer so much larger than 23?

In the first problem any pair of guests can match, and 23 guests make $\binom{23}{2} = 253$ pairs. In the second, only the m guest–host pairs count. To get the same number of chances of a match we need $m = 253$, and this is exactly why 253 appears in both answers.

Checks

- Exact products confirm both answers. The true probability of a shared birthday is 0.4757 for 22 guests and 0.5073 for 23. The true value of $1 - (364/365)^m$ is below $\frac{1}{2}$ for $m = 252$ and is 0.5005 for $m = 253$.
- $\frac{253}{365} = 0.69315\ldots$ and $\ln 2 = 0.69315\ldots$, agreeing to five decimal places, so the given approximation does not affect either answer.
- A simulation of 200 000 parties of 23 gave a shared birthday 50.7% of the time.

Common slips

- Computing the probability that two *particular* guests share a birthday, or adding $\frac{1}{365}$ once for each pair. Overlapping events can't simply be added; working with the complement avoids this.
- Using $\frac{364}{365}$ for every factor in the first product. The $(r+1)$th guest must avoid r days, not just one.
- Including the host in the count of guests for the last part. The host is a separate person, so 253 guests are needed.

Beyond STEP

The heuristic here is that a match among k objects in N boxes becomes likely once k is about $\sqrt{2N\ln 2} \approx 1.18\sqrt{N}$. It underlies "birthday attacks" in cryptography, which find collisions in an N-bit hash after about $2^{N/2}$ tries rather than 2^N. Behind the approximation is the Poisson limit: the number of matching pairs is roughly Poisson with mean $\binom{k}{2}/365$, and $\mathrm{P}(\text{no matches}) \approx \mathrm{e}^{-k(k-1)/730}$. Blitzstein and Hwang, *Introduction to Probability*, discusses the birthday problem and its Poisson approximation in detail.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.